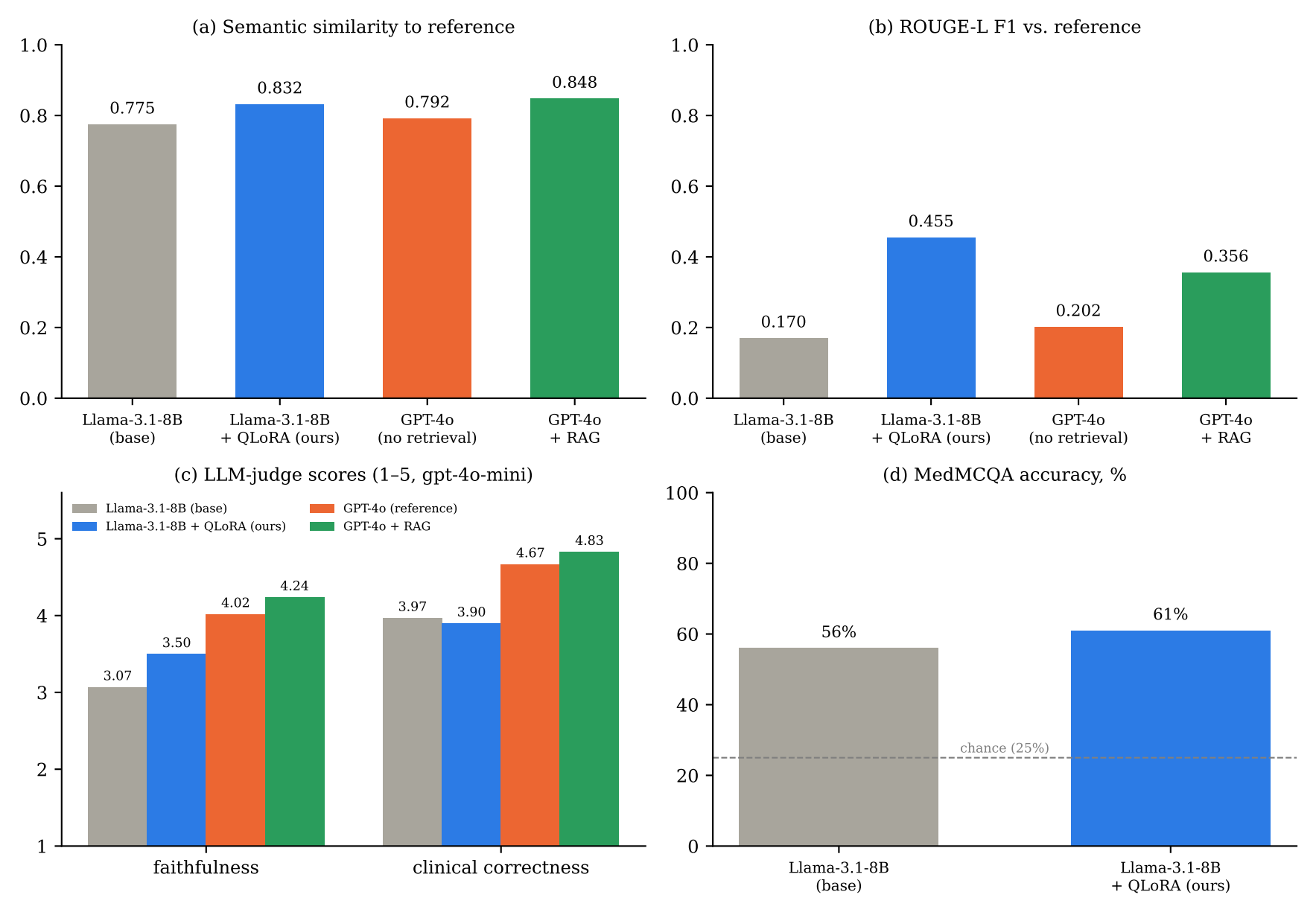
(a) Semantic similarity to reference
1.0
0.8
0.6
0.4
0.2
0.0
0.775
0.832
0.792
0.848
Llama-3.1-8B (base)
Llama-3.1-8B + QLoRA (ours)
GPT-4o (no retrieval)
GPT-4o + RAG
(b) ROUGE-L F1 vs. reference
1.0
0.8
0.6
0.4
0.2
0.0
0.170
0.455
0.202
0.356
Llama-3.1-8B (base)
Llama-3.1-8B + QLoRA (ours)
GPT-4o (no retrieval)
GPT-4o + RAG
(c) LLM-judge scores (1–5, gpt-4o-mini)
Llama-3.1-8B (base)
Llama-3.1-8B + QLoRA (ours)
GPT-4o (reference)
GPT-4o + RAG
5
4
3
2
1
3.07
3.50
4.02
4.24
3.97
3.90
4.67
4.83
faithfulness
clinical correctness
(d) MedMCQA accuracy, %
100
80
60
40
20
0
56%
61%
chance (25%)
Llama-3.1-8B (base)
Llama-3.1-8B + QLoRA (ours)